

Female Infertility

Based on Pharmacotherapy Handbook, 7th Edition

Prepared for Pharmacy Yr IV UG Students



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Definition & Epidemiology

Female infertility: inability to conceive after 12 months of regular unprotected intercourse.

Primary vs. secondary infertility.

Affects ~10% of reproductive-age women globally.



Primary vs. secondary infertility

- 1. Primary Infertility
 - A woman has **never achieved a pregnancy** after:
 - **12 months** of regular unprotected sexual intercourse (if <35 years), or
 - **6 months** (if ≥35 years).
 - Ž She has **never conceived before.**



Primary vs. secondary infertility...

- **2. Secondary Infertility**
 - A woman is **unable to conceive again** after previously having:
 - A **confirmed pregnancy**,
 - Regardless of whether it resulted in live birth, miscarriage, ectopic pregnancy, or termination.
 - Ž She **has conceived before**, but now cannot



Etiology of Female Infertility

Ovulatory dysfunction (e.g., Polycystic Ovary Syndrome (PCOS), hypothalamic dysfunction).

Tubal damage (Pelvic Inflammatory disease (PID), endometriosis).

Uterine factors (fibroids, adhesions).

Endocrine causes (thyroid disorders, hyperprolactinemia).



Ovulatory Dysfunction vs. Tubal Damage

Feature	Ovulatory Dysfunction (PCOS, Hypothalamic)	Tubal Damage (PID, Endometriosis)
Definition	Failure or irregular ovulation leading to infertility	Blockage or damage to fallopian tubes impairing fertilization
Common Causes	PCOS, hypothalamic dysfunction, obesity, stress	Pelvic inflammatory disease, endometriosis, prior pelvic surgery
Clinical Clues	Irregular cycles, anovulation, hirsutism, obesity	History of PID, pelvic pain, infertility despite regular cycles
Diagnosis	Hormonal assays (FSH, LH, AMH), ultrasound, ovulation tracking	HSG (hysterosalpingography), laparoscopy, pelvic imaging
Management	Lifestyle modification, ovulation induction (clomiphene, letrozole), ART if needed	Antibiotics for infection, surgical correction, In Vitro Fertilization (IVF) for severe damage



Uterine Factors vs. Endocrine Causes

Feature	Uterine Factors (Fibroids, Adhesions)	Endocrine Causes (Thyroid, Hyperprolactinemia)
Definition	Structural abnormalities of the uterus interfering with implantation	Hormonal imbalances affecting ovulation and menstrual cycles
Common Causes	Fibroids, intrauterine adhesions (Asherman's), congenital anomalies	Hypothyroidism, hyperthyroidism, elevated prolactin levels
Clinical Clues	Abnormal bleeding, recurrent miscarriage, infertility	Menstrual irregularities, galactorrhea, systemic thyroid symptoms
Diagnosis	Ultrasound, hysteroscopy, MRI	Thyroid function tests, serum prolactin, pituitary imaging if needed
Management	Surgical removal (myomectomy, adhesiolysis), ART if persistent	Thyroid hormone replacement, dopamine agonists (cabergoline, bromocriptine)



Diagnostic Evaluation (General)

- Detailed history and physical examination.
 - Ovulation assessment: progesterone level, ovulation predictor kits.
 - Imaging: pelvic ultrasound.
 - Tubal patency: HSG (hysterosalpingography).
 - Hormonal profile: FSH, LH, TSH, prolactin, AMH.



Reading Assignment

- FSH, LH, TSH, prolactin, AMH.
 - Function, clinical use, and interpretation

General Management Overview

- Management depends on etiology.
 - Ovulation induction: clomiphene citrate or letrozole (first-line).
 - Correcting hormonal abnormalities (thyroid, prolactin).
 - Surgical options for endometriosis, fibroids.
 - Assisted reproductive technology (ART).



Updated Gynecology Recommendations (ACOG/ESHRE 2023–2024)

Letrozole
recommended as
first-line for
ovulation induction
in PCOS.

Lifestyle
modification
strongly advised for
women with obesity.

Early referral to ART
after age 35 if no
conception within 6
months.

Avoid unnecessary
laparoscopy unless
indicated (e.g.,
severe
endometriosis).



Pharmacologic Therapy

Letrozole: higher live birth rates than clomiphene (PCOS).

Gonadotropins for second-line induction when oral agents fail.

Metformin in women with PCOS and insulin resistance.

Cabergoline for hyperprolactinemia.



1-Letrozole (aromatase inhibitors)

Dose: 2.5–7.5 mg orally once daily for 5 days

Start day: Day 3–5 of menstrual cycle

Dose adjustment: If no ovulation → increase by 2.5 mg each cycle (max 7.5 mg)

Notes:

- First-line for ovulation induction in **PCOS**
- **Higher live birth rates** than clomiphene
- Stop after ovulation is confirmed or after 5–6 cycles



2. Gonadotropins

- Used as **second-line** when oral agents (letrozole/clomiphene) fail.

a. Recombinant FSH / hMG

- **Starting dose: 50–75 IU SC daily**
- **Adjust every 3–7 days** based on follicular response
- **Typical range: 75–150 IU/day**
- **Goal: Controlled follicular development (1–2 dominant follicles)**



2. Gonadotropins ...

- b. Trigger Shot
(Ovulation Induction)
 - hCG 5,000–10,000 IU
IM/SC
or
 - r-hCG (Ovidrel) 250
mcg SC



3-Metformin (for PCOS with insulin resistance)

Initial: 500 mg once daily with meals

Titration: Increase by 500 mg every 1–2 weeks

Typical therapeutic dose: 1,500–2,000 mg/day

- Split: **500 mg TID** or **1,000 mg BID**

Notes:

- Improves ovulation in insulin-resistant PCOS
- Useful in combination with letrozole when metabolic features are prominent



4. Cabergoline (for Hyperprolactinemia)

Dose: 0.25–0.5 mg twice weekly

Titrate gradually based on prolactin levels

Maximum typically 2 mg/week (higher in resistant cases)

Notes:

- Preferred over bromocriptine due to fewer side effects
- Normalization of prolactin restores ovulation



Quick Clinical Pearls

Letrozole is **first-line** for anovulatory infertility in PCOS.

If no ovulation after 3–6 cycles → escalate dose or switch to **gonadotropins**.

Metformin is most beneficial in **overweight/obese PCOS** with insulin resistance.

Always evaluate **thyroid function, prolactin, and male factor** before starting ovulation induction.

Monitor with **follicular ultrasound** to reduce risk of multiple gestation or Ovarian Hyperstimulation Syndrome (OHSS).



Assisted Reproductive Technologies (ART)

In vitro fertilization (IVF).

Intracytoplasmic sperm injection (ICSI).

Ovarian stimulation protocols for IVF.

Success rates depend on age and ovarian reserve.



Complications & Risks

Ovarian hyperstimulation syndrome (OHSS).

Multiple gestation with ovulation induction.

Emotional and financial burden.



References

- ACOG Practice Bulletin on Infertility, 2023.
- ESHRE Guidelines on Female Infertility, 2024.
- Pharmacotherapy Handbook, 7th Edition (general endocrine principles).

